

JOSHUA VINU KOSHY

joshuvk16@gmail.com | +91 8657406860 | Navi Mumbai, India | LinkedIn | Github

Education

Fr. C. Rodrigues Institute of Technology (FCRIT)

Bachelor of Engineering in Computer Engineering | **GPA: 9.69/10**

Navi Mumbai, India

2022 – 2026

Experience

Project Intern | InfinityPool Finnotech Pvt. Ltd.

Jul 2024 – Present

- Designed a **Backtrader-powered trading engine** with multi-timeframe simulations, optimized data loaders, and real-time market feeds, enabling faster, production-ready strategy evaluation for equity portfolios.
- Architected **agentic AI workflows** with multi-agent orchestration and long-term memory, enabling autonomous equity monitoring and portfolio optimization for live portfolios.

Full Stack Developer Intern | FCRIT

Jan 2025 – Jun 2025

- Led a complete revamp of **fcrit.ac.in** with **React.js** UI redesign, admin dashboards, and backend migration from CakePHP to a **Node.js + MySQL** microservice-style architecture, reducing load time by 40%.
- Implemented secure REST APIs with JWT auth, RBAC, and caching for exams, faculty management, and feedback modules, improving reliability and maintainability of the portal.

Projects

Signal Agnostic Backtesting Framework | *React.js, Node.js, PostgreSQL, Python, Backtrader, Pandas, NumPy, Selenium, Chartink Screener*

Jun 2025

- Engineered a **backtesting framework** accommodating **multiple trading strategies** and **15+ custom indicators** through optimized event-driven execution and asynchronous data handling.
- Engineered **optimized data loaders** with chunked OHLCV ingestion, memory-mapped datasets, and async caching, improving backtest throughput by 35–40% & also implemented risk and performance parameters(Sharpe, max drawdown, win rate, volatility, equity curves) to compare trading strategies.

Real-Time Location-Based Event Platform | *React.js, Node.js, Open-StreetMap, Firebase*

Feb 2025

- Built a real-time event sharing system with **geolocation tracking** and **async media uploads**, reducing map update latency to under 300ms using **OpenStreetMap/Leaflet** and cached POI search.
- Developed a secure backend with **Google OAuth**, **Firestore real-time listeners**, and **Node.js APIs**, achieving consistent sub-100ms read/write operations.

Technical Skills

Languages & Core Skills: Python, JavaScript (ES6+), TypeScript, Java, C/C++, SQL

Web Technologies: React.js, Next.js, Node.js, Express.js, Flask, FastAPI, REST APIs, WebSockets

AI/ML & LLMs: TensorFlow, Keras, scikit-learn, Generative AI (Gemini), LangChain, RAG pipelines

Databases & Storage: MySQL, MongoDB, PostgreSQL, Firebase, Supabase, Vector DBs (FAISS, Chroma)

Cloud, DevOps & Tools: Git, GitHub, AWS, Docker, CI/CD (GitHub Actions), Vertex AI, Netlify, Vercel

Leadership & Voluntary Experience

- Head, AI & Deep Learning Club** – Organized 10+ technical events for 250+ participants on campus.
- Speaker** for AI/ML workshop at **Cryptex 2024**, trained 30+ students.
- Guest speaker** at Short Term Training Program (Dept. of IT), delivered AI/ML sessions for 100+ attendees.

Publications & Achievements

- Research paper: “**Fraudulent Transaction Detection in Reverse Auction Systems Using Machine Learning and Blockchain-Assisted Anomaly Analysis**” accepted at **ICNTE 2026**.
- 3rd Position** at **TechSprint 2026** (Open Innovation Hackathon), organized by **Google Developer Group on Campus GSSSIETW**.
- Second Runner-Up** at **Hacktivate 2024** (AI + Cybersecurity hackathon), organized by Hacktify.
- Secured **first rank** in the third year of Computer Engineering with a perfect **GPA of 10.0**.
- 1st Prize** at **Cryptex 2023** (Web Development) and **3rd Prize** at **CodeMania**, a competitive programming contest organized by the Computer Society of India.